Do Longer Work Hours Reduce Happiness and Impair Mental Health?

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Topic Selection Rationale

- In Japan, issues surrounding long working hours and oppressive work environments (“black companies”) are widely discussed
- Youth suicide remains a leading cause of death in Japan, representing a critical societal challenge

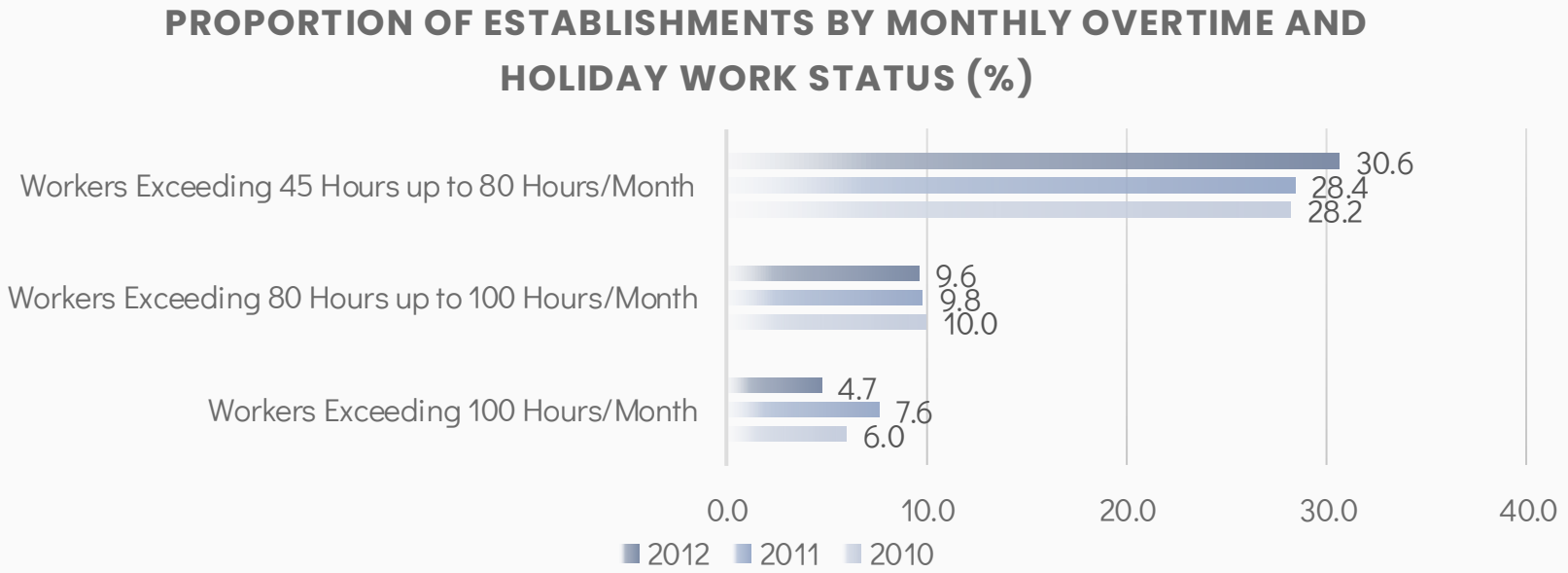
Objective

Investigate the empirical relationship between working hours, subjective happiness, and mental health outcomes

COUNTRY	SUICIDE RATE 2023 (PER 100K) ↓	MALE SUICIDE RATE 2023 (PER 100K)	FEMALE SUICIDE RATE 2023 (PER 100K)
Greenland	75.57	108.21	39.10
Suriname	28.69	46.21	11.66
South Korea	28.13	39.93	16.26
Guyana	25.36	42.69	8.34
Lithuania	24.76	43.30	8.42
Russia	24.28	42.87	8.06
Uruguay	22.49	38.58	7.74
Japan	21.46	30.13	13.27
Ukraine	21.28	37.66	7.33
Cuba	21.13	34.32	8.07
Lesotho	20.80	35.44	6.91
Guam	19.93	35.28	4.17
Taiwan	19.14	26.42	12.03

Overtime Work Exposure (2012 MHLW Survey)

- >100 hours/month: 4.7% of surveyed establishments reported employees exceeding 100 overtime hours per month
- 80-100 hours/month: 9.6% of establishments reported employees working 80-100 overtime hours per month
- 45-80 hours/month: 30.6% of establishments reported employees working 45-80 overtime hours per month



Mandatory Physician Consultation System

A statutory health intervention designed to detect early signs of fatigue and mental health distress caused by overwork, and aimed to prevent severe physical and mental health conditions

Purpose

Assess workers' health for high-risk individuals with accumulated fatigue, provide medical guidance, and mandate workplace follow-up actions

Consultation Scope

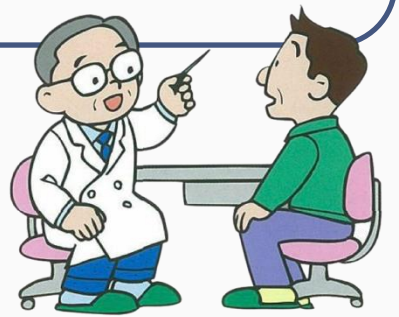
Evaluation of working hours, cumulative fatigue, sleep quality, and overall physical and mental health.

Target Audience

- Workers exceeding 80 hours/month of overtime or work during holiday with self-reported fatigue (requires employee request)
- Mandatory employer-led implementation for all worker exceeding 100 hours/month of overtime

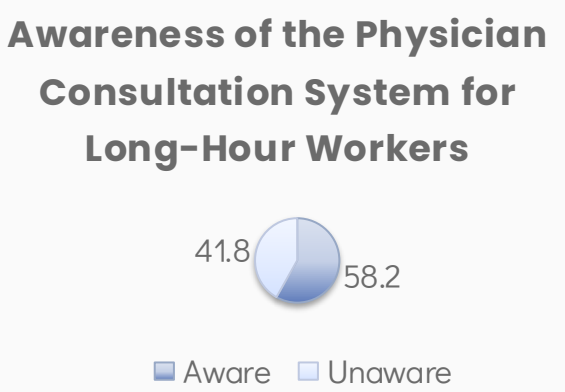
Post-Consultation Measures

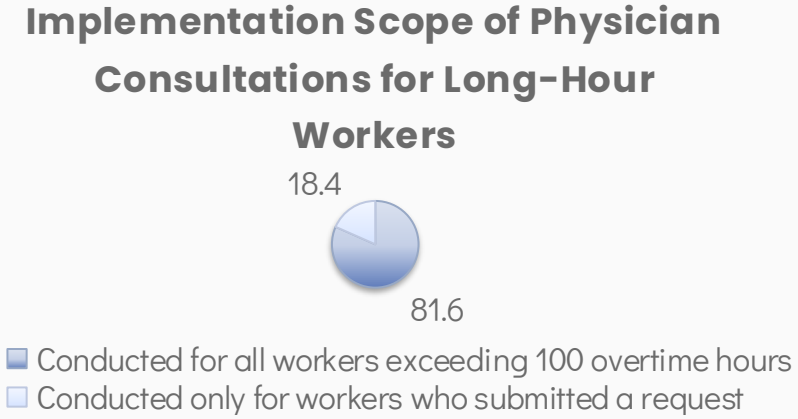
Employers are legally required to implement corrective actions (e.g., reducing work hours, altering job responsibilities, or placing restrictions on work duties) based on physician feedback



Mandatory Physician Consultation System

- The physician consultation system acts as a protective mechanism for long-hour workers
- As of 2012, approximately 40% of establishments were unaware of the system, indicating insufficient institutional diffusion
- Implementation standard among active establishments:
  - 81.6% applied mandatory consultations to all workers exceeding 100 overtime hours
  - 18.4% restricted consultations strictly to workers who explicitly requested them
- Major Concern: Expanding program awareness and ensuring consistent & proactive application across all businesses



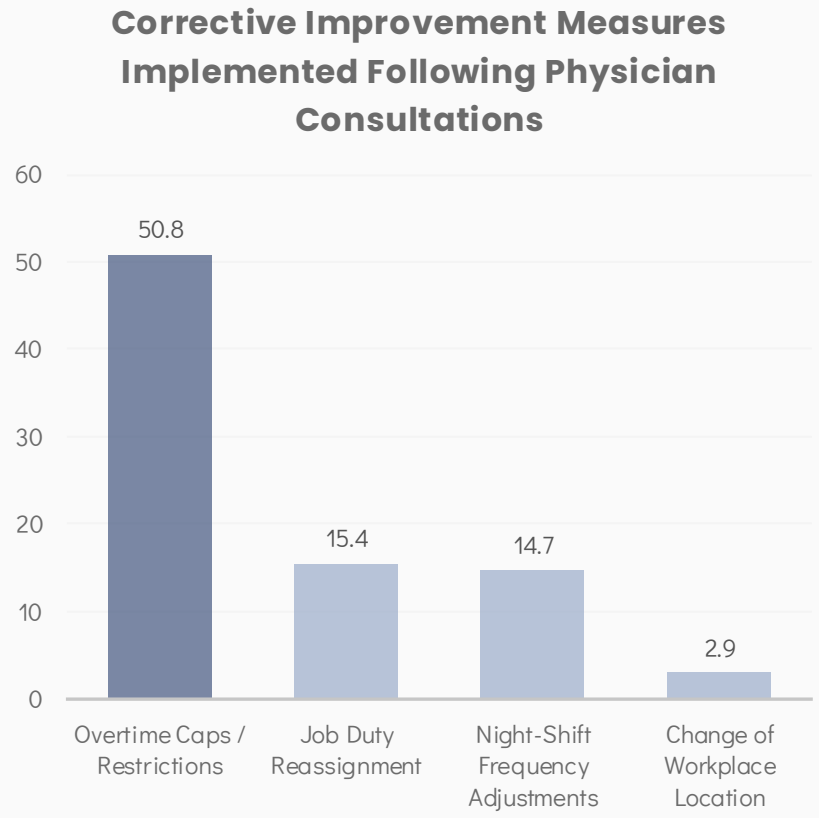


Mandatory Physician Consultation System

- 82.9% of establishments conducting physician consultations implemented corrective workplace adjustments.
- Primary adjustments taken:
  1. Overtime Caps / Restrictions: 50.8%
  2. Job Duty Reassignment: 15.4%
  3. Night-Shift Frequency Adjustments: 14.7%

The "Karoshi" Threshold (Ministry of Health, Labour and Welfare)

- Overtime exceeding 100 hours in a single month or an average of 80 hours/month over 2–6 months
- Exceeding this line significantly increases the incidence of cerebrovascular/cardiovascular events and major depressive disorders
- Physician guidance prioritizes immediate work hour reductions for workers with 80 hours of monthly overtime corresponds to working until near the last train every single day



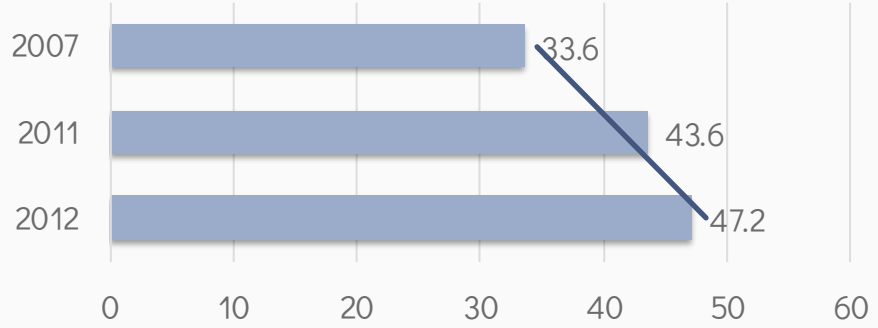
Corporate Mental Health Initiatives

Workplace Program Adoption Trends

- 47.2% of establishments actively conduct mental health initiatives (up from 2011)
- Adoption exceeds 90% in large organizations with 300+ employees

Primary Workplace Mental Health Initiatives

- Employee Training & Information Dissemination: 46.7%
- Manager & Supervisory Training: 44.7%
- Internal Mental Health Consultation Services: 41.0%





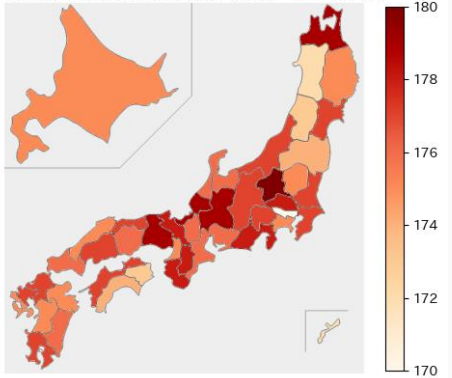
Long Hours, Compensation Claims, and Happiness

- Mental disorder claims rose from 927 (2008) to 1,257 (2012) highlighting the growing psychological toll of overwork
- The Easterlin Paradox: Income increases subjective well-being only up to a specific point, beyond which happiness gains plateau
- Long working hours erode key non-monetary determinants of happiness, especially personal time, physical health, and interpersonal relationships
- Excessive work hours impair both health outcomes and fundamental life satisfaction

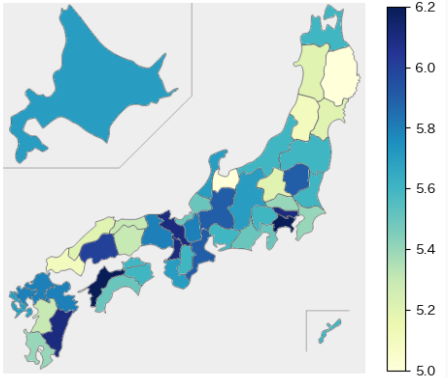
Category / Year		2008	2009	2010	2011	2012
Brain/Heart Disease Claims	Total	889	767	802	898	842
	Resulted in Death	304	237	270	302	285
Mental Disorder Claims	Total	927	1136	1181	1272	1257
	Resulted in Death	148	157	171	202	169

Domestic Analysis: Work Hours vs. Life Satisfaction

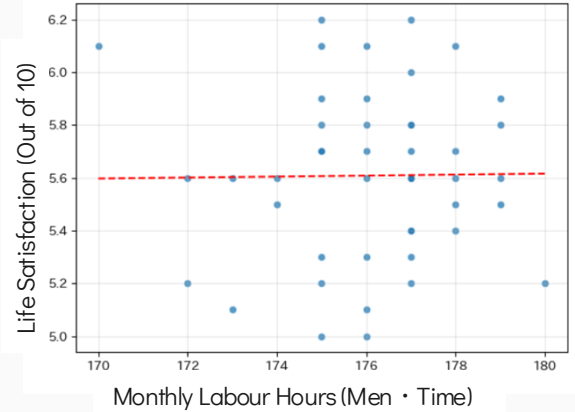
Monthly Actual Working Hours by Prefecture (Men, 2024)



Life Satisfaction by Prefecture (Out of 10, 2024)



Labour Hours & Life Satisfaction (by Prefecture) correlation=0.01



- No statistically significant correlation was identified between male monthly working hours and life satisfaction across Japanese prefectures (r=0.01)
- Tokyo (shortest working hours) reported high satisfaction, whereas Gunma (longest working hours) reported lower satisfaction

Analytical Insight

- Unlike international trends, working hours within Japan are relatively uniform across regions
- Domestic life satisfaction is driven more strongly by local socio-economic, environmental, and lifestyle factors rather than work hours alone

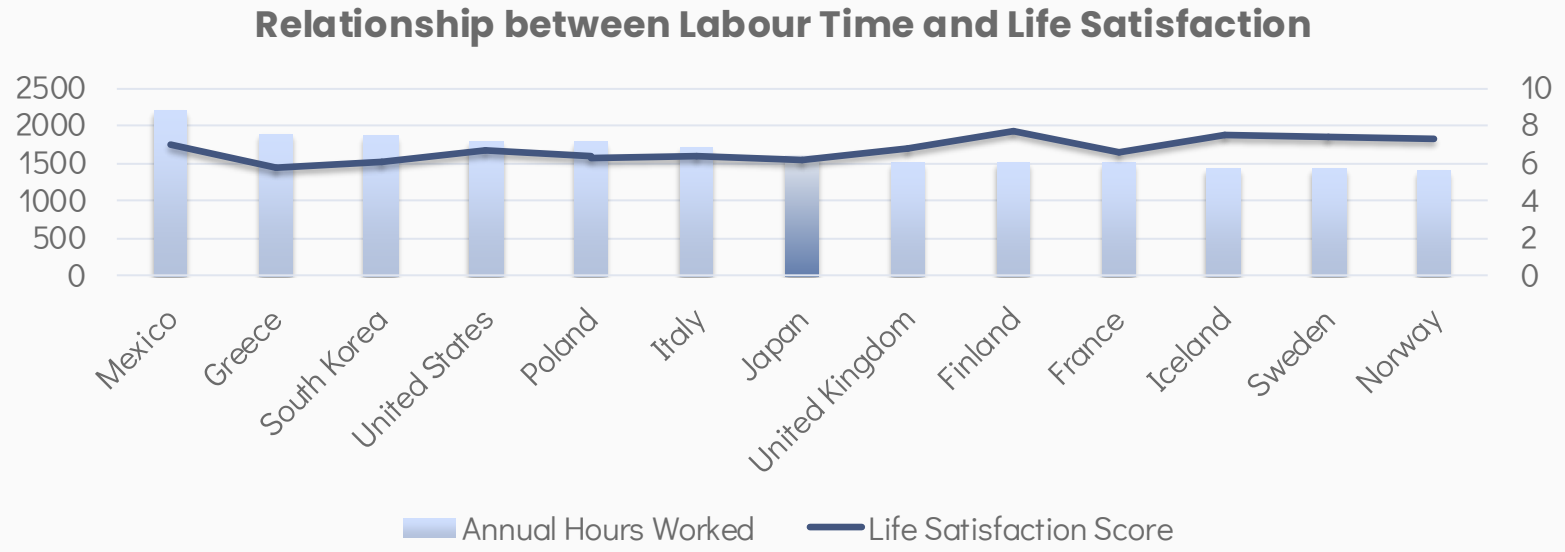
International Comparison: Work Hours vs. Happiness

Global OECD & World Happiness Report Data Summary

- International data reveals a negative relationship
- Nations with longer average working hours generally exhibit lower life satisfaction.

Country	Annual Hours Worked	Life Satisfaction Score
Mexico	2193	6.98
Greece	1898	5.78
South Korea	1865	6.04
United States	1796	6.72
Poland	1785	6.37
Italy	1709	6.42
Japan	1617	6.15
United Kingdom	1512	6.73
Finland	1509	7.74
France	1491	6.59
Iceland	1434	7.52
Sweden	1431	7.35
Norway	1407	7.26

- Japan's average annual hours (1,617 hrs) appear moderate due to a high proportion of non-regular/part-time workers (~40%) meanwhile full-time employees work significantly longer hours
- Japan's life satisfaction score (6.15) remains among the lowest across developed economies
- Countries with short working hours (e.g., Nordic nations) benefit from higher per capita income and robust social safety nets
- Long working hours exert a negative impact on both health and well-being, supporting the core hypothesis



Conclusion & Strategic Implications

Conclusion

Long working hours are believed to have a negative impact on both health and well-being, and the chosen topic, “Do longer working hours lead to lower levels of well-being and negatively affect mental health?” was generally supported

- Overtime exceeding the Karoshi threshold (>80–100 hrs/month) remains prevalent, driving increased mental health claims (1,257 in 2012)
- Interventions like physician consultations are expanding, but an awareness gap (~40% unawareness rate) hinders universal protection
- Domestically in Japan, work duration is a key contributing factor, alongside broader social factors such as local socio-economic, environmental, and lifestyle factors.
- Internationally, shorter work hours strongly correlate with higher happiness
- Addressing work-life quality requires reducing work hours, establishing compliance with legal health mechanisms, and reforming organizational culture

Methodology & Data Workflow

Gathered official statistics and different survey data. Then, used a variety of tools and techniques for analysis and visualization depending on the goal.

Data Collection

- Statistics on working hours, workplace accidents, and mental health were obtained from the Ministry of Health, Labor and Welfare and the Ministry of Internal Affairs and Communications' e-Stat "Social Life Statistical Indicators"
- Annual working hours by country were obtained from the OECD
- Happiness levels by country were obtained from the World Happiness Report 2025
- Life satisfaction by prefecture was obtained from the PDF version of Sony Life's "2024 Survey on Attitudes Toward Life by Prefecture (All 47 Prefectures)."

Excel

Compiled data from the OECD and the World Happiness Report to create a chart showing "working hours" and "life satisfaction" by country

Methodology & Data Workflow

Gathered official statistics and different survey data. Then, used a variety of tools and techniques for analysis and visualization depending on the goal.

Google Colab

Utilized Python (japanmap, matplotlib) to plot color-scaled regional heatmaps of prefectural work hours and life satisfaction scores. Executed linear regression models (numpy, matplotlib) to compute scatter plots and correlation coefficients

Script for Japan Map Visualization by Prefecture

```
import matplotlib.pyplot as plt
from matplotlib.colors import Normalize
import japanize_matplotlib
from japanmap import picture

def draw_map(data, title, cmap_name):
    cmap = plt.get_cmap(cmap_name)
    norm = Normalize(vmin=min(data.values()), vmax=max(data.values()))
    colors = {p: tuple(int(c*255) for c in cmap(norm(v))[:3]) for p, v in data.items()}
    fig, ax = plt.subplots(figsize=(6, 7))
    ax.imshow(picture(colors)); ax.axis('off'); ax.set_title(title, fontsize=14)
    fig.colorbar(plt.cm.ScalarMappable(cmap=cmap, norm=norm), ax=ax, shrink=0.7)
    plt.show()
```

```
draw_map(work_hours,   '都道府県別 月間実労働時間数（男性・2024年度）', 'OrRd')
draw_map(satisfaction, '都道府県別 生活満足度（10点満点・2024年）',   'YlGnBu')
```

Script for Displaying Correlation in a Scatter Plot

```
import numpy as np
prefs = list(work_hours.keys())
x = [work_hours[p] for p in prefs]
y = [satisfaction[p] for p in prefs]
r = np.corrcoef(x, y)[0, 1]

fig, ax = plt.subplots(figsize=(7, 5))
ax.scatter(x, y, s=40, alpha=0.75, edgecolor='white')
z = np.polyfit(x, y, 1)
xs = np.array([min(x), max(x)])
ax.plot(xs, z[0]*xs + z[1], '--', color='red')
ax.set_xlabel('月間実労働時間数（男性・時間）')
ax.set_ylabel('生活満足度（10点満点）')
ax.set_title(f'労働時間と生活満足度（都道府県別）  相関 r = {r:.2f}')
ax.grid(alpha=0.3)
plt.show()
print(f'相関係数 r = {r:.3f}')
```

References & Data Sources

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